NGUYEN TRONG DANG-KHOA

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CAREER OBJECTIVE

A highly **motivated IT fresh-graduate** with a strong foundation in **software development** and a keen interest in **Artificial Intelligence**. Eager to apply **problem-solving** and **quick learning skills** to build **innovative, robust, and scalable solutions** in any field. Seeking a **Intern/Fresher or higher level** position where I can contribute to the development of high-performance products, leveraging my experience with **AI/ML**.

EDUCATION

Ho Chi Minh City University of Industry and Trade *2021 – 2025*
Bachelor of Information Technology | GPA: 3.23/4

AWARDS AND HONORS

- Encouragement Award for "Emergent products and topics in the information technology field", HUIT *7/2023*
- First prize in "Young coder's Competition: Finding the Best and Brightest", HUIT *6/2023*
- Third prize in "Science Fair" *11/2020*

SKILLS

Languages: Python, Javascript, C++

Frontend: ReactJS, HTML/CSS, Bootstrap

AI/ML: PyTorch, LangChain, Ollama

Testing: Pytest, Locust, Playwright, Selenium

Tools: Git, Copilot, VSCode, Postman

Backend: FastAPI

Databases: MySQL, MongoDB, Redis

DevOps: Linux, Docker, GitHub Actions (CI), Bash

Soft skills: Teamwork, Research skills, Growth mindset

English: TOEIC(Listening & Reading) 720 / IELTS overall 6.0

Certifications: khoa15.github.io/certifications.html - IELTS/ TOEIC/ Awards

PROFESSIONAL EXPERIENCE

Backend Intern | EdunetJSC *09/2024 – 12/2024*

Tech Stack: Python, FastAPI, Docker, Linux, MQTT, ONNX

- Optimized the Frigate **object-detection** pipeline, improving accuracy by **2%** through **data-augmentation** and **hyperparameter tuning**.
- Assisted in the maintenance of IoT water quality monitoring stations, gaining practical experience in a small-scale production environment.

PROJECTS

SIML RAG – Retrieval-Augmented Generation System *Personal Project*

github.com/Khoa15/simpl-RAG | Locust Performance Report

Tech Stack: Python, FastAPI, React, LangChain, GeminiAPI, Redis, Celery, Docker, Pytest, Locust

Problem & Objective: Recognizing that existing RAG systems often suffer from high latency and resource inefficiency, I aimed to design and implement an efficient, scalable RAG system capable of handling multiple concurrent users while minimizing response time and resource consumption.

Result: Developed a multi-instance RAG system using Python, FastAPI, and LangChain. Integrated the Gemini API for text generation. Implemented a custom caching mechanism and request pooling to reduce median query latency to less than 300ms.

My Role & Responsibilities: Solely responsible for the system's design, from backend development with FastAPI to the implementation of caching and resource management. Also developed an auto-delete feature for inactive users.

Exci – AI-Powered English Learning App

University Project

github.com/Khoa15/exci.git | Team Lead (Score: 9.5/10)

Tech Stack: Flutter, ASP .NET, Winform .NET, Ollama, Python

Problem & Objective: To address the lack of personalized and engaging tools for English learners, I aimed to build an AI-driven mobile application that enhances the vocabulary learning experience through a chatbot and spaced repetition functionality.

Result: Developed a learning app with a chatbot AI and spaced repetition functionality. Integrated Ollama models for the chatbot and vocabulary search. The backend was built using ASP.NET to manage user data, achieving chatbot response times of 0.5–1s.

My Role & Responsibilities: As Team Lead, I was responsible for the backend development, integrating the AI models, and ensuring the seamless operation of the user data management system. Managed team workflow and performed code reviews.

Note-Taking App with OCR

University Project

github.com/Khoa15/dury.git | Team Lead (Score: 9.8/10)

Tech Stack: Spring, MongoDB, Firebase, Android (Java), Flask

Problem & Objective: Recognizing that converting text from images into editable notes is a manual and time-consuming process, I aimed to streamline this by integrating a highly accurate Optical Character Recognition (OCR) system into a comprehensive note-taking application.

Result: Developed a comprehensive note-taking application with a robust Spring Boot backend. We created a dedicated Flask microservice for Optical Character Recognition (OCR) using the Tesseract library, which demonstrated a high degree of precision, achieving approximately 92% accuracy when digitizing text from printed sources.

My Role & Responsibilities: Led the team, designed the backend architecture, and was directly responsible for developing the Flask microservice for OCR. Also managed the team's workflow and conducted code reviews.

Age Classification with CNN

University Project

github.com/Khoa15/huit-ai.git | Team Lead (Score: 9.0/10)

Tech Stack: Python, React native, Tensorflow, Flask

Problem & Objective: To address the complexity of accurately classifying human age from facial images, I aimed to build a complete, end-to-end computer vision system that demonstrates proficiency in model training and full-stack integration.

Result: Trained a CNN model on the UTKFace dataset, achieving a test set accuracy of 40%. Built a Flask server to handle classification requests and integrated it with a React Native client, creating a complete end-to-end system.

My Role & Responsibilities: Responsible for leading the team, training and fine-tuning the CNN model, and building the Flask server to serve the model. Demonstrated the ability to develop a full-stack AI application from research to deployment.